When should text and images be clustered together?

Zhang and Leung (2025)
Joint Text-and-Image Clustering for Social Science Research

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Computational Social Science with Images and Audio

Which representation, and when is fusion worth it?

Which representation strategy best clusters paired text and images, and **when** is joint clustering worth using?

Joint methods helped most where separate text and image clusters **barely agreed**; where they agreed more, simple methods did about as well.

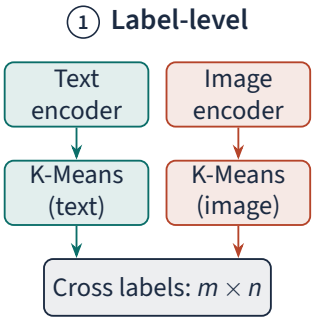
A conclusion from two examples, not an established general law.

MY READING

A useful decision guide, more than a new algorithm.

Shown: less fragmentation. Not yet shown: more faithful measurement of what a pair means.

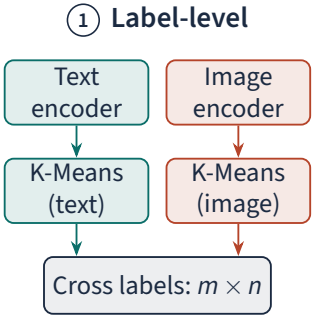
Three workflows differ in where text and image meet



Two separate partitions; every text × image label pair is a cell ($5 \times 5 = 25$)

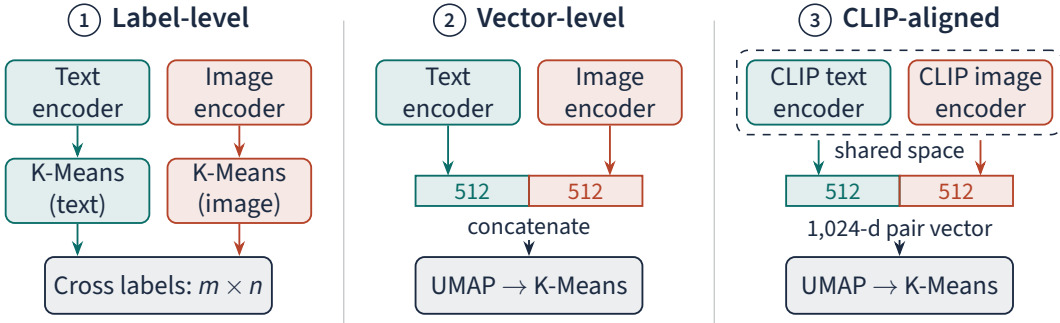
Adapted from Zhang and Leung (2025), Figure 1 (p. 9)

Three workflows differ in where text and image meet



Adapted from Zhang and Leung (2025), Figure 1 (p. 9)

Three workflows differ in where text and image meet



Same clustering step: what changes is where the modalities meet

Adapted from Zhang and Leung (2025), Figure 1 (p. 9)

Two contrasting corpora, four different kinds of evidence

336,921

CASM-China: Weibo text–image pairs on protests, 2010 to mid-2017

A post's text is repeated for each of its images

1,297

BU-NEmo: U.S. gun-violence news, headline–image pairs from 21 outlets

(A) AGREEMENT

AMI

Do text-only and image-only partitions agree?

(B) GEOMETRY

S_Dbw

Compact and separated? Lower is better

(C) COVERAGE

Pruning loss

Pairs lost keeping only the largest clusters

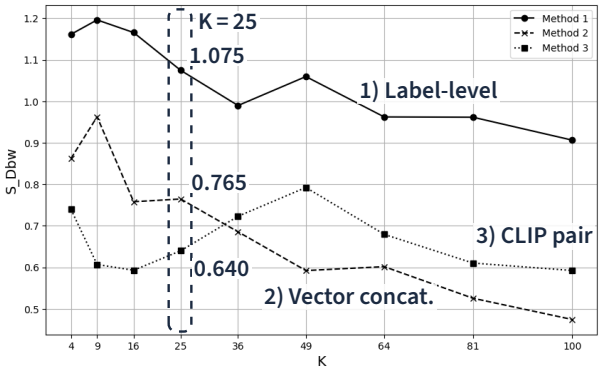
(D) MEANING

Human coding

Share of 10 posts on the modal theme (not reliability)

Class 1a–1b: CNN/ViT and CLIP for *supervised* measurement (ad partisanship).
Here: the same encoders for *unsupervised discovery*.

Joint fusion wins on geometry, not on human coherence



Original exhibit: Figure 7, p. 28 (names, values, K = 25 marker added). S_Dbw by K, CASM, K-Means; lower is better.

CASM-CHINA RESULTS

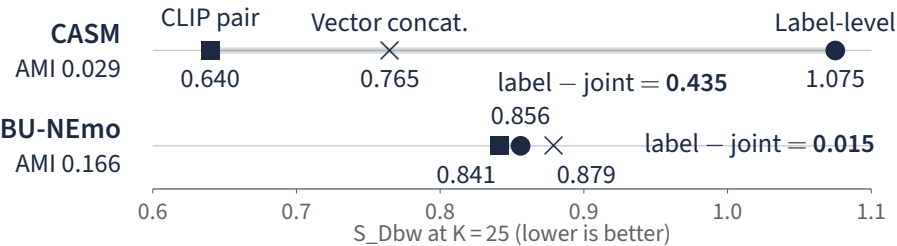
AMI = **0.029**: text and image partitions nearly independent

AT K = 25	S_Dbw ↓	Human ↑
1 Label-level	1.075	**0.820**
2 Vector concat.	0.765	0.808
3 CLIP pair	**0.640**	0.724

The ranking flips; across K = 20–30 coders show no stable winner.

Separate K = 100 test (10 × 10 cells): the 25 largest label cells leave out **over half** of all pairs.

BU-NEmo's gain is small, yet more than AMI differs



ALSO DIFFERS	CASM-China	BU-NEmo
Content	Chinese protest posts	English gun-violence news
Producers	Weibo users	journalists at 21 outlets
Size	336,921 pairs	1,297 pairs
Encoders	multilingual text + Places365; CLIP	OpenAI text + Google Vertex

Suggestion: hold 1 dataset and encoders fixed, vary caption-image pairings in controlled ways from reinforcing to conflicting to noise and see how AMI, S_Dbw, and hand-coding responds

Joint methods avoid splintering, but are not more meaningful

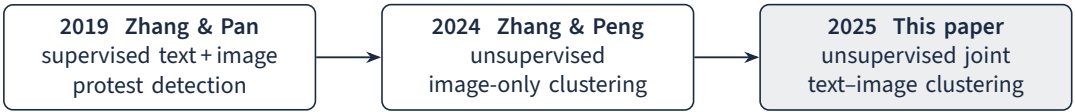
SAME 25 CLUSTERS, OPPOSITE RANKINGS



Shape = S_Dbw (how compact and separate the clusters are). Humans = do the posts share a theme? The tidiest clusters made the least sense to people.

1. **Easy opponent.** Joint methods beat the crudest way to combine text and image clusters, not the best merging methods eg. late-fusion, consensus (not tested)
2. **Disagreement is ambiguous.** Text and image clusters that don't line up may add different information, or may just conflict, be noisy, or reflect a weak model
3. **Untested settings.** We aren't told how text and image were balanced, and the rule for choosing the number of clusters is never varied
4. **Counting and coding.** A post with 9 images counts 9 times, and no one checks whether two coders would agree

The contribution is integration and guidance, not invention



ESTABLISHED BEFORE 2025

- Multi-view clustering, late fusion (2017–2024)
- CLIP's shared image–text space (2021)
- Multimodal topic models (2022)
- AMI, S_Dbw, human topic checks

ADDED HERE

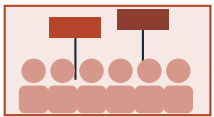
- A head-to-head comparison on two corpora
- One reusable evaluation workflow
- Code and hardware notes for sociologists

Strong as an accessible methodological bridge; thinner as an algorithmic contribution.

Chao et al. 2017; Radford et al. 2021; Liu et al. 2021; Zosa and Pivovarova 2022; Zhou et al. 2024.

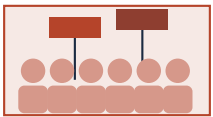
Suggestion: test if clusters keep the caption–image relation

= Reinforcement



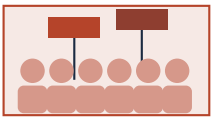
“People gather for a peaceful demonstration.”

+ Complementarity



“Workers demand unpaid wages at the rally.”

≠ Contradiction



“Violent extremists attack the city.”

Hypothetical: one drawn crowd, three invented captions; not real images, news or paper findings.

Validate	Independent coders label sampled pairs by relation and report reliability; then test whether the clusters separate those relations.
Use with care	Suits visual-framing studies of protest coverage. Descriptive only: not causal evidence or grounds for sanctions; protesters can be re-identified.

FOR DISCUSSION

1. if text and image clusters don’t agree, does that mean they’re telling complementary stories, or could the two messages conflict? How could we tell?

2. Is an accessible, reproducible synthesis a sufficient methodological contribution when the underlying fusion and validation tools already exist?